



Software Quality Assurance Plan

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1. INTRODUCTION

1.1 Objectives

Scope: The STP will include detailed test coverage of the all-encompassing Digital Accessibility System for Sanad, a product that helps people live with disabilities. We will be testing all primary functions of our platform, including user management, interactive map features, community forum posts, service reservations, information on learning and jobs, assistive technology interoperability, and super admin moderation capabilities. The STP details both the functional and non-functional requirements of the product by validating specific functional requirements (for example, user login and registration; navigation through maps; posting messages and replies in forums) and non-functional requirements (for example, performance of the system; security of user information; and compliance with WCAG 2.1 AA standards for accessibility).

The STP will test the Digital Accessibility System using a layered architecture approach and will perform testing at five different levels: unit, integration, system, acceptance, and UAT. Repetitive tasks will undergo automated tests (for instance, API validation), whereas manual tests will occur primarily on the user interface components and assistive technology features. Risk-based testing will be conducted on high-risk areas such as data security and real-time services (e.g., video chats).

Our testing team will have several members that include developers, QA engineers, and accessibility experts. The tools we will use for our testing include automated UI testing tools (Selenium), automated API testing tools (Postman), and automated unit testing tools (Jest). The Digital Accessibility System will be tested in three environments: development, staging, and production. Other key dependencies include the Google Maps API and third-party payment processors which we will need for testing our Digital Accessibility System's integration capabilities.

Schedule: Test Integration within a Development Timeline:

Testing activities will align with the overall project development timelines. Testing will start during the early phases of development by performing unit tests and continue through integration testing, followed by system testing after integration has occurred and final acceptance testing (or User Acceptance Testing-UAT) before the product is released. The major milestones for the testing effort will be the completion of the unit tests on or before the end of week 8, completion of integration tests on or before the end of week 12, completion of system tests on or before the end of week 16, and completion of UAT by the end of week 18. The total testing effort is estimated to require 200 person-hours, and the budget is allocated to cover tools and training.

Products to be delivered by the testing effort: The testing products will include test cases, test scripts, defect reports, summaries of test results, and matrices that tie the testing back to SRS and SDS requirements.

Major work activities within the Testing effort:

The major work activities that will be completed within this effort are the development of the test cases, execution of the tests, logging of defects, performing regression testing, and validating the

fixes. During the course of this effort, the majority of the activities will focus on testing for access (such as screen reader accessibility), and the majority will also focus on security (such as testing for encrypted communications).

Key Milestones for the Testing effort:

Unit test completion (week 8), Integration test approval (week 12), System test sign-off (week 16), User Acceptance Testing completion (week 18), and Final Release readiness (week 20).

Budget and Funding Requirements for the Testing effort:

The estimated budget for the testing effort is \$50,000; and the breakdown of the testing effort is: 40% Unit/Integration Tests, 30% System Tests/Acceptance Tests, 20% Regression Testing, and 10% Documentation

1.2 Testing Strategy

Objective: Testing on the Sanad platform ensures compliance with the SRS functional requirements (e.g. Integrity & Security of User Registration etc., Accessible Map Interactions) and Design specifications of the SDS (e.g. RESTful API and Layered Architecture) while identifying defects and verifying that Users with Disabled Access have access to the most usable version of the finished product. Furthermore, Defution will confirm that the Sanad conforms to various standards such as WCAG 2.1 AA and PDPL.

Items Required for Testing Purposes: Users will be able to test all Core Components of Sanad, including Administrative Dashboards, User management Sub-systems, Maps & Accessible Places, Forums, Service Booking, and Profile Management Systems. External Integrations such as Google Maps and Payment Gateways will also be tested for their Reliability.

Feature Requirements for Testing: All Functional Requirements of the SRS will be needed to be tested (FR1-FR6), including User Authentication, Map Filtering and Navigation, Forum Moderation, Interpreter Booking, and Browsing Assistive Technology. The Non-Functional Aspects will include Performance (Response Time less than 3 seconds), Security (HTTPS Encryption) and Accessibility (Screen Reader support).

Items that are not related to testing are hardware integrations (user created hardware designs), internal components of a third-party vendor (internal code from Google Maps) and post-launch operational aspects (user training).

Management and the Technical Approach - Use a risk-based approach and automated tools for increased efficiency. Utilize black-box testing (to test the user interface) white-box testing (to test the backend logic) and exploratory testing (to ensure accessibility). Test environments will be replicas of production environments (replicating AWS cloud setup).

To determine whether tests pass or fail, all requirements outlined in the SRS must be met without a critical defect for a test to be considered a pass. Failure will be determined by a 5% excessive transaction failure rate, response time equal to or greater than 3 seconds and/or anything not compliant with WCAG; if acceptance criteria are met, the test case pass rate must be equal to or exceed 95%.

Individual Roles and Responsibilities - The QA Lead manages QA testing and makes strategy decisions. Developers complete their own unit tests. QA Engineers manage Integration/System Testing and handle WCAG compliance for Accessibility Testing. The Project Manager coordinates the schedules of all test team members.

Milestones - As described in the objective sections of this document, by Week 8 to complete all Unit Testing.

Schedule - Based on project phases and must complete Regression Testing within 24 hours after a fix has been made.

Risk Assumptions and Constraints - Risks include integration issues with third-party services; the assumption is that external APIs will be stable and one of the constraints is that the project will only have a limited number of Accessibility expertise available and a budget to purchase tools.

1.3 Scope

The STP will be updated during scheduled releases (Version 1.1) and will also be updated in the event of an emergency (critical defects). Change management procedures outlined in Section 4 of the SRS will be used for all updates. All updates will be submitted to the Change Control Board (CCB) for approval.

Updates will be sent to stakeholders via email. Stakeholders can obtain a version-controlled copy of their STPs in either GitHub or GitLab.

Git will provide version control and configuration management for STP versioning. Each STP version is tracked with a log of all changes made (the initial STP version 1.0, all future versions and subsequent changes). The configuration management for each of the associated STPs follows the constraints defined for each technology stack, such as React, Node.js and MySQL, all of which are directly related to the SDS.

1.4 Reference Material

- Project Authorization: SRS Purpose has implied Project Authorization through the use of Digital Solutions for people with disabilities.
- Project Plan: SDS Structure will provide a structure for the development of the Sanad Project (Development Plan)
- Quality Assurance Plan: Will align with non-functional requirements defined in SRS (performance, security) and will follow the guidelines set out in the SRS.
- Configuration Management Plan: Will align with the constraints outlined in the SDS Section 3.2.
- Organizational Policy and Procedures: SRS Section 1.3: definitions and acronyms; SDS Section 1.3
- Associated Standards: IEEE 830-1998, ISO/IEC/IEEE 29148:2018; WCAG

2.1; PDPL, CRPD, WHO reports (referenced in SRS).

- Supplemental Reference Materials: SRS (Version 1.0), SDS (Version 1.0); Test Plan (SDS Reference) for Sanad.

1.5 Definitions and Acronyms

Terminologies are listed in the table below:

Terminology	Definitions
Sanad	Sanad Platform is a unified digital system that provides accessible digital services to empower people with impairments.
Accessibility	The process of designing products, devices, services, or environments that can be accessed and used by people with disabilities.
Architectural Design	Architectural design generally refers to the high-level design of a software system along with how the various components of the system interact with each other.
Assistive Technology	Assistive technology (AT) may refer to some tools or devices that support people with disabilities in performing tasks.
Backend services	Backend services describe the server-side components of a software system.
Business logic	Business logic includes the rules and processes that determine how data is processed or how the software system behaves.
The client-server model	The client-server model, in general, describes a model of a distributed application where the client (web/mobile app) requests services from the server (backend).
Component design	Component design generally refers to the detailed design (i.e., the structure, state, behavior, etc.) of each of the components of a software program.
Data Flow	The route data follows across a system, from being input, processed, and output.
Data Model	The representation of the type data structures and how they relate to each other in a system.
Digital Accessibility	Providing access to digital tools and platforms for people with disabilities.
Frontend	The end user or client facing portion of a software application or engagement with it, including the user interface and user interactions.
Functional Requirements	Details on technical specifications of what the system has to do for functionality - i.e. user registration, map engagements, etc. Derived from the SRS, and answered in the SDS.

Implementation Design	The detailed implementation plan, including choice of technology stack hired and use of other services and systems.
Integration Points	The locations where the system connects to, or is related to, an external service or system.
Layered Architecture	Layers were through and functional application of design principles and patterns creating the software architecture for layering.
Microservices	A type of software architecture in which applications are built as a set of small, independent services.
Modular System Architecture	A design paradigm in which the system is partitioned into independent modules, to facilitate activity and maintenance.
Non-Functional Requirements	The nonfunctional specifications of how the system performs (e.g., security, scalability, accessibility).
RESTful APIs	APIs that adhere to principles of REST for communication between layers of the system.
Scalability	The ability of a system to accommodate the growth of users or data.
Security Protocols	An actions taken by a system that delivers protection against a threat.
Self-Determination	The ability of individuals to choose, make decisions, and have control over their lives.
Social Inclusion	The process of improving the conditions under which individuals can participate in society, derived from the WHO
Technology Stack	The combination of technologies utilized to aid in the construction of a system for consumers.
User Interface (UI) Design	Within computing, UI design pertains to the real and/or visual and interactive design elements that consumers engage with.
WebSocket	A network communication protocol for real-time bidirectional data exchange.

Table 1- Definitions

Acronyms	Meanings
SDS	Software Design Specification
SRS	Software Requirements Specification a document outlining the software's objectives, features, and limitations.
SQL	Structured Query Language
2FA	Two-Factor Authentication (a security process that enables users to provide two separate forms of verification to validate their identity).

API	Application Programming Interface (a collection of rules and protocols to access a software application or web-based tool). AWS: Amazon Web Services (a cloud computing platform offered by Amazon).
CRPD	Convention on the Rights of Persons with Disabilities (a United Nations treaty to protect the rights of persons with disabilities).
DGA	Digital Government Authority (a Saudi Arabian government agency focused on digital transformation and accessibility).
Flutter	Flutter is a user interface toolkit for building applications which are compiled natively (a mobile app development open-source framework by Google).
HTML5	HTML5 or Hypertext Markup Language 5 is the latest version of the markup language for web pages.
HTTPS	HTTPS, or Hypertext Transfer Protocol Secure is a secure version of HTTP designed for secure communication over computer networks.
IEC	IEC, or International Electrotechnical Commission, is a standardization organization in the field of electrotechnology. IEEE: IEEE, Institute of Electrical and Electronics Engineers, is an organization of professionals that advances technology for humanity in the field of electrical engineering and computing.
ISO	ISO stands for International Organization for Standardization and is the international standard-development organization.
JWT	JSON Web Token (a compact, URL-safe method for representing claims to be transferred between two parties).
MySQL	My Structured Query Language (an open-source relational database management system).
Node.js	A JavaScript runtime built on Chrome's V8 JavaScript engine (a server-side

	development platform).
React	A JavaScript library for building user interfaces (a library for creating interactive web user interfaces).
Redis	Remote Dictionary Server (an in-memory data structure store used as a database, cache, and message broker).
REST	Representational State Transfer (an architectural style for designing networked applications, typically used for APIs).
UI	User Interface (the space where human and machine interactions occur).
UN	United Nations (an international organization that was formed in 1945 to promote peace and cooperation).
W3C	World Wide Web Consortium (a worldwide community that focuses on building open standards for the Web)..
WCAG	Web Content Accessibility Guidelines (guidelines for improving the accessibility of web content for people with disabilities).
WHO	World Health Organization (a United Nations agency that specializes in international public health).
XSS	Cross-Site Scripting (a sort of vulnerability seen in web applications).
AT	Assistive technology may refer to some tools or devices that support people with disabilities in performing tasks
etc.	abbreviation for the Latin phrase "et cetera," which means "and other things" or "and the rest". It is used at the end of a list to indicate that there are more similar items that could be included but are not explicitly mentioned.

Table 2- Acronyms

2. TEST ITEMS

2.1 Program Modules

User Management and Authentication subsystem: Create an account, log in and create/update profile as well as change password, create 2FA, validate JWT tokens and role-based access for Clients, Volunteers and Admin (Admin Needs to be able to test all users); as such, unit test the backend logic of each function and integration tests with database via MySQL.

Map and Accessible Places Subsystem: This subsystem should have tests for rendering maps, searching/filtering places, submitting and reviewing places. Validate that the Google Maps API works with accurate GPS tracking, and verify that the accessibility features (ADA accessible wheelchair routes) are functioning. All functionality will have automated testing via scripts simulating API calls, along with manual "eyeball" checking of all UI overlays.

Community and Forum Subsystem: An example of functionality within this subsystem would be testing creating a new post, moderating comments and reporting content. Validate that the post is safe for public viewing via notifications and Category Filters. Performance Testing of the forums will be performed under high traffic conditions, as well as Accessibility Testing of all posts by screen readers.

Service and Appointment Booking Subsystem: This subsystem will have tests for booking interpretation services or volunteer services, tracking status of bookings and video call capabilities. Validate how well the matching algorithm is executed, real-time status updates (for endorsement only), statistical reporting, and Encryption of all services. Test integration with WebSocket events and Security Testing of encryption of end-to-end communications.

Profile and Accessibility Preferences Subsystem: Test all switches used by a user, including high contrast (dark mode) text size preference and history. Validate adaptations made to the user interface based on user-set preferences, and confirm persistence of user data/history.

Admin Dashboard and Moderation Subsystem: This subsystem will provide all Admins with the ability to perform reports and analytics functions to review the activity of users and the content of users on the site. Validate all role permissions granted by the Admin to each user and validate the integrity of all data exported from the system to users. All Admin Dashboard items must be tested for security and access control.

2.2 Job Control Procedures

Backend Processing includes Testing Tasks Scheduled to Run (Notifications Sent); Database Data Backups; API Sync via Google Maps. Ensure that when Real-Time Updates are sent via WebSocket Events for other Devices, all Events are sent in the correct order. Finally, simulate production loads via Automatically Created Scripts to confirm that Automatic Processes do not create a Deadlock Condition.

Data Operations include CRUD in MySQL with Indexing for Better Performance; Data Redundancy via Data Replication; Audit Logging via Triggering; Data Integrity via Constraints.

Integration with 3rd-Party Payments Integration (PayPal) and File Storage Services (AWS S3) are tested to ensure the transaction process is secure, and that Errors are handled correctly.

2.3 User Procedures

A documentation test will validate the accuracy, completeness and accessibility of all user guides, tutorials, and help sections (WCAG compliance). A document will include a step by step instruction to test how users can use the site. Usability tests should be done with people with disabilities to ensure that instructions are understandable and easy to follow.

Onboarding materials will include testing of registration flows, profile setup instructions, and tutorials for accessing accessibility settings. Testing to confirm the availability of multilingual support (English/Arabic) and error messages that are clear.

2.4 Operator Procedures

For production support, tests will be run on production monitoring tools and processes, deployment scripts and help desk processes. Testing for system recovery (MTTR <10 mins) and failover. Testing when a system goes down will include checks for alerts.

For all maintenance procedures, tests should confirm the process of updating, backing up and recovering after a disaster. Testing may include compliance with the operator's access control and logging requirements (PDPL) and the stability under simulated high-load scenarios

3. FEATURES TO BE TESTED

3.1 Functional roles to be tested

3.1.1 Login and Account Management

- Login via email and password.
- Register as a new user.
- Forgot password.
- Update the account details.
- Two-step verification (if available).
- Manage permissions (user/administrator).

3.1.2 Interactive Map of Accessible Places

- View accessible places.
- Filter places by accessibility type.
- View details and ratings.
- Search for a specific location.
- Integration with Google Maps.

3.1.3 Forum Section

- Create new posts.
- Comment and reply.
- Edit and delete private posts.
- Report abusive content.
- Administrator tools (delete – ban – approve).
- View categories and trending topics.

3.1.4 Service Request System Book a Sign Language Interpreter

- Specify the date and time.
- Confirm your appointment.
- Cancel or reschedule.
- Integration with visual tools.
- Bookings Administrator Dashboard

3.1.5 Learning and Careers Section

- View training courses.
- View and apply for jobs.
- Filter by type.
- Access assistive technology resources.

3.1.6 Assistive Technology Support

- Screen reader support.
- Keyboard navigation.
- Text of zoom.
- High contrast mode.
- ARIA labels mode in items.

3.2 *Non-functional features to be tested*

3.2.1 Performance

- Page load time ≤ 3 seconds.
- API response time ≤ 1.5 seconds.
- Scaling-up stress test.
- Real-time connection quality.

3.2.2 Security

- HTTPS encryption.
- API protection using JWT.
- Protection against SQL injection, XSS, and CSRF attacks.

- Secure password storage.
- Secure session management.

3.2.3 Accessibility

Compliant with WCAG 2.1 AA standard and includes:

- Alternative text for images
- Color contrast.
- Keyboard navigation.
- Screen reader support.
- Clear templates and appropriate error messages.

3.2.4 Usability

- Ease of user navigation.
- Clarity of interfaces.
- Design consistency according to SDS.

3.3 Integration Tests

- Integration with Google Maps API.
- Integration with payment systems (if applicable).
- Integration with WebSocket for instant connectivity.
- Front-end and back-end integration via RESTful APIs.
- MySQL database operations.

3.4 System-Level Testing

- Support for various devices (mobile, tablet, laptop).
- Support for various browsers.
- System compatibility with the architecture designed in SDS.
- Error management.
- Event logging and monitoring.

4. FEATURES NOT TO BE TESTED

4.1 *Features Outside the Current Release*

These features are not listed in the current project requirements in the SRS or the SDS, and therefore will not be tested:

- Proposed future features for subsequent expansions.
- Any additional augmented reality (AR) or virtual reality (VR) services.
- Support for assistive devices connected via Bluetooth or external hardware.

4.2 *Third-Party Components*

No internal functionalities of external service providers will be tested, as these systems are managed by independent companies:

- The internal code of the Google Maps API will be tested, but integration will be the only focus.
- The internal systems of electronic payment providers (if applicable).
- The internal architecture of email or OTP messaging services.

4.3 *Outside the Project Scope (Out-of-Scope)*

Elements not included in the project scope :

- User training or development of user training manuals.
- Post-release operations.
- Operational oversight.
- Advanced AI-powered analytics .

4.4 *Functions that do not require testing*

Some functions do not require testing because they are performed outside the system:

- Government registration procedures or linking national ID cards.
- Document verification through official channels.
- Activities that rely on real data from government agencies not integrated into the current version.

4.5 *Tests Not Included*

Tests not included in the current testing plan:

- Advanced Penetration Testing.

- Extremely High-Performance Testing (Load Testing above Expected Usage Levels).
- AWS Security Testing (Cloud Infrastructure Security Testing).

4.6 Excluded Technical Components

Technical elements present in the design but not directly tested:

- Internal servers, hosting system, networks, firewall.
- The MySQL database engine is a standalone program; queries will only be tested through the system.

4.7 Reasons for Exclusion

These elements were excluded for the following reasons:

- They are not required in the current release according to SRS.
- They depend on a third party and cannot be tested internally.
- They do not affect the quality of the current release.
- They require special environments or permissions that are not available.
- They will be tested in future expansions.

5. APPROACH

5.1 Component Testing

Component testing verifies the validity of individual components (React UI components, Node.js services, MySQL operations, accessibility tools, and WebSocket handlers) before platform build integration.

5.1.1 Authentication & Account Management

5.1.1.1 Login with Email & Password

Field	Description
Test ID	SANAD_LOGIN_001
Feature	User login via email & password (FR1 – User Authentication)
Prerequisite	User account exists in the database with valid email & hashed password.
Test Data	1) Valid email & password 2) Valid email & wrong password 3) Empty email 4) Empty password 5) Invalid email format (user@email.com)
Test Procedure	1. Navigate to Login page. 2. For each data set (1–5): enter email & password. 3. Click “ Login ” button.
Expected Result	1) User is redirected to the correct Home dashboard according to role (User / Volunteer / Admin). 2–5) Display accessible error message (with ARIA role="alert") “Email or password is incorrect”, focus moves to error message, no login occurs.
Pass/Fail Criteria	Pass if all negative cases are rejected and the valid case logs in correctly without exposing sensitive information.

Table 3 - Login with Email & Password

5.1.1.2 Registration of New User

Field	Description
Test ID	SANAD_REGISTER_001
Feature	New account registration (FR1)
Prerequisite	Registration page is accessible; email not already registered.
Test Data	a) All fields valid b) Missing required field c) Weak password d) Duplicate email e) Invalid phone number format
Test Procedure	1. Open Register page. 2. Fill and submit form with each test data set.
Expected Result	a) Account created, confirmation message shown, user may be redirected or sent verification email b) Inline validation messages per field: “This field is required”. c) Password strength warning (must contain X chars, digits, special char). d) Message: “This email is already registered”. e) Message: “Phone number must follow national format (05XXXXXXXXX)”.

Accessibility Check	All error messages are announced by screen readers; fields have proper labels and error associations.
Pass/Fail Criteria	Pass if validation rules and messages match SRS & SDS and data is stored correctly in the DB.

Table 4- Registration of New User

5.1.1.3 Forgot Password / Reset Flow

Field	Description
Test ID	SANAD_FORGOT_PASSWORD_001
Feature	Password reset via email (FR1)
Prerequisite	User account exists; email server configured.
Test Data	1) Valid existing email 2) Non-existing email 3) Invalid format / empty email
Procedure	1. Click “ Forgot Password ”. 2. Enter email as per data set. 3. Submit request and check email inbox. 4. Click reset link and set new password.
Expected Result	1) Message: “If this email exists, a reset link has been sent.” Email received with one-time token, reset succeeds and old password no longer valid. 2–3) Same generic message (no user enumeration), no email sent for non-existing emails.
Security Criteria	Token is single-use, time-limited, and stored hashed in DB; link expires after use or timeout.

Table 5- Forgot Password / Reset Flow

5.1.1.4 Two-Factor Authentication (2FA)

Test ID	SANAD_2FA_001
Procedure	Enable 2FA in profile → login again → system asks for OTP → enter correct, wrong, expired, or empty OTP.
Expected Result	Correct OTP → login success; wrong/expired/empty → deny access with accessible error message.

Table 6- Two-Factor Authentication (2FA)

5.1.2 Profile & Accessibility Preferences

5.1.2.1 Update Profile & Accessibility Settings

Field	Description
Test ID	SANAD_PROFILE_001
Feature	Profile update + accessibility preferences (high contrast, text size, language).
Prerequisite	Logged-in user; existing profile.
Test Data	a) Valid profile info, valid preferences b) Missing required fields c) Invalid email / phone d) Toggle all accessibility options

Procedure	1. Open Profile page. 2. Edit fields and toggle accessibility options. 3. Save changes. 4. Refresh page and re-login on different device.
Expected Result	Data saved successfully; accessibility settings applied immediately (theme change, font size, etc.) and persist across sessions/devices. Invalid inputs show correct validation messages.
Accessibility Criteria	Controls are keyboard navigable; ARIA attributes used for toggles; contrast meets WCAG 2.1 AA.

Table 7- Update Profile & Accessibility Settings

5.1.3 Map & Accessible Places Component

5.1.3.1 Searching & Filtering Accessible Places

Field	Description
Test ID	SANAD_MAP_SEARCH_001
Feature	Map search & filters (FR2 – Accessible Map)
Prerequisite	Map page is accessible; Google Maps API key configured; DB contains sample places.
Test Data	• Keyword: “hospital”, “mall” • Filter: wheelchair, braille, ramps • Invalid search string (“@@@”)
Procedure	1. Open Map. 2. Enter search keyword and apply filters. 3. Use keyboard only to move focus between results and map pins.
Expected Result	Map centers on relevant region; list shows matching accessible places; invalid search leads to “No places found” message without breaking the UI.

Table 8-Searching & Filtering Accessible Places

5.1.3.2 Submitting a New Accessible Place

Test ID	SANAD_MAP_SUBMIT_002
Procedure	Open “Suggest Place” → fill mandatory fields (name, type, accessibility options, coordinates) → submit with valid, then incomplete, then invalid coordinates.
Expected Result	Valid → place saved in pending list for admin moderation; Incomplete/invalid → inline accessible validation errors; no DB insertion.

Table 9- Submitting a New Accessible Place

5.1.4 Forum & Community Component

5.1.4.1 Create Post & Comment

Field	Description
Test ID	SANAD_FORUM_POST_001
Feature	Create forum post & comments (FR3 – Community)
Prerequisite	Logged-in user; network available.
Test Data	a) Valid title + content b) Empty content c) Post containing prohibited words (to trigger moderation)
Procedure	1. Navigate to Forum . 2. Click “ New Post ”. 3. Submit each test data set. 4. Add a comment to a post.
Expected Result	a) Post created and visible in list b) Error “Content is required” c) Post flagged for moderation / hidden per SDS rules, comment successfully added.

Table 10-Create Post & Comment

5.1.4.2 Report Abusive Content

Test ID	SANAD_FORUM_REPORT_002
Procedure	Click “Report” on a post → choose reason → submit.
Expected Result	Confirmation message; admin dashboard shows new report with all metadata (user, time, reason, link to post).

Table 11-Report Abusive Content

5.1.5 Service & Appointment Booking Component

5.1.5.1 Book Sign Language Interpreter

Field	Description
Test ID	SANAD_BOOKING_INTERPRETER_001
Feature	Booking interpreter (FR4 – Service Booking)
Prerequisite	Registered user; interpreters available at the selected time slot.
Test Data	a) Valid date/time, service type, notes b) Date/time in the past c) Missing mandatory fields
Procedure	1. Open Service Booking . 2. Choose “Sign Language Interpreter”. 3. Fill booking form using each dataset and submit.
Expected Result	a) Booking created; confirmation email/notification sent, status “Pending/Confirmed”. b) Error “Cannot book in the past”. c) Error for each missing field.

Table 12-Book Sign Language Interpreter

5.1.5.2 Cancel / Reschedule Booking

Test ID	SANAD_BOOKING_MODIFY_002
Procedure	From “My Bookings”: select existing booking → cancel → confirm; then try reschedule with valid/invalid new times.
Expected Result	Status updates correctly (Cancelled/Rescheduled) and corresponding notifications sent; logs created for audit.

Table 13-Cancel / Reschedule Booking

5.1.6 Admin Dashboard & Moderation

5.1.6.1 Manage Users & Roles

Field	Description
Test ID	SANAD_ADMIN_USERS_001
Feature	Admin managing user accounts & roles
Prerequisite	Logged in as Admin; at least 3 sample users.
Procedure	1. Open Admin Dashboard → Users. 2. Change role (User → Volunteer, Volunteer → Admin) with valid/invalid operations (removing last Admin). 3. Deactivate a user, then try to log in as that user.
Expected Result	Valid changes persist and appear in UI, system prevents removing last active Admin; deactivated users cannot log in and receive appropriate message.

Table 14-Manage Users & Roles

5.1.6.2 Approve/Reject Places, Posts and Bookings

Field	Description
Test ID	SANAD_ADMIN_CONTENT_002
Procedure	Use Admin panel to review pending accessible places, flagged posts, and bookings; approve some, reject others with reasons.
Expected Result	Approved items visible to users; rejected items remain hidden, with reason stored in DB for audit; notifications sent.

Table 15-Approve/Reject Places, Posts and Bookings

5.2 Integration Testing

Integration testing checks that combined components (UI + API + DB + external services) behave as a cohesive system.

5.2.1 User Journey – New User to Completed Booking

Field	Description
Test ID	SANAD INT JOURNEY 001
Prerequisite	Staging environment with external services (Google Maps, payment gateway, email, WebSocket) configured.
Scenario	New user registers → confirms email → logs in → sets accessibility preferences → browses map → books interpreter → receives notifications.
Steps	1. Complete registration & email verification. 2. Login. 3. Update accessibility preferences. 4. Open Map & choose accessible place. 5. Book interpreter. 6. Check notification (web & email).
Expected Result	All systems interact correctly: account created, preferences saved, map data loaded, booking stored in DB, notification sent, and no broken links or missing data across modules.

Table 16-User Journey – New User to Completed Booking

5.2.2 WebSocket Integration – Real-Time Updates

Field	Description
Test ID	SANAD_INT_WEBSOCKET_002
Scenario	Admin updates booking status → user receives real-time update.
Steps	1. User logs in from Device A (browser tab). 2. Admin logs in from Device B. 3. Admin changes booking status (Pending → Confirmed).
Expected Result	Device A shows real-time status change without page refresh; WebSocket connection remains stable; fallback to polling if WebSocket unavailable.

Table 17-WebSocket Integration – Real-Time Updates

5.2.3 Payment Integration Scenario (If Enabled)

Field	Description
Test ID	SANAD INT PAYMENT 003
Scenario	User books paid service → completes payment through external gateway.
Expected Result	Payment form redirects to gateway; after success, user returns to Sanad with booking marked “Paid”; on cancel, booking remains unpaid or cancelled with proper message; Sanad never stores raw card data.

Table 18-Payment Integration Scenario (If Enabled)

5.2.4 Admin & Reporting Integration

Field	Description
Test ID	SANAD_INT_ADMIN_004
Scenario	Changes in bookings, new places, reports, and forum posts are reflected in Admin analytics page.
Expected Result	Dashboards show accurate counts & charts based on the latest DB data; filters work by date, type, and user role.

Table 19-Admin & Reporting Integration

5.3 Conversion Testing

Conversion testing includes verifying that all required data elements and historical records in the Sanad Platform are correctly converted when moving from an older environment to a new system format.

The goal is to ensure that user accounts, accessibility preferences, bookings, map locations and forum posts remain accurate and intact after conversion, with no loss or corruption.

Technique:

Data is exported converted using migration scripts and re-imported into the new environment. The development team reviews record counts and validates formats to ensure complete accuracy.

Completion criteria:

All converted data is fully correct, accessible and approved by the project supervisor.

5.4 Job Stream Testing

Job stream testing ensures that the Sanad Platform operates correctly within a production ,like environment and that all automated background processes run as expected.

This includes notification jobs, scheduled database operations and system monitoring tasks.

Technique:

The development team executes scheduled tasks and observes their behavior to confirm that they run in the correct order and produce the expected outputs.

Completion criteria:

All scheduled jobs must complete successfully without errors or delays and final approval is given by the supervisor.

5.5 Interface Testing

Interface testing ensures that the Sanad Platform operates efficiently and correctly with all external interface systems.

This includes integrations with Google Maps API, OTP/email services, payment systems and WebSocket connections.

Technique:

Interfaces are demonstrated to the project supervisor to confirm that all client required functionalities are working properly and that external systems exchange correct data with Sanad.

Completion criteria:

All interfaces must be fully approved and performing without errors.

5.6 Security Testing

Security testing ensures that all system control mechanisms and auditability features in the Sanad Platform are functional and protect user data.

This includes authentication, authorization, encryption and protection from common vulnerabilities.

Technique:

The team performs controlled tests such as invalid login attempts, token misuse, SQL injection simulation and access control checks.

Completion criteria:

All security controls work as intended and all results are validated and approved by the project supervisor.

5.7 Recovery Testing

Recovery testing ensures that the Sanad Platform can restart properly after failures and that backup and recovery features operate as designed.

This confirms that the system can return to normal operation without losing data.

Technique:

Controlled failures are simulated (server shutdown, database interruption) and the team tests the recovery tools to ensure full restoration.

Completion criteria:

The system must restart successfully, retrieve backed-up data and operate normally.

5.8 Performance Testing

Performance testing ensures that the Sanad Platform meets expectations for response time, availability, scalability and overall speed.

Technique:

Load and stress tests are performed to verify page load time, API performance and behavior under increased user traffic.

Completion criteria:

System pages must load within the required response time and meet the performance criteria defined in the SRS.

5.9 Regression Testing

Regression testing ensures that any updates or changes applied to the Sanad Platform do not negatively affect previously working functionality.

Technique:

Previously executed test cases are re-run after every fix or enhancement to verify system stability.

Completion criteria:

All previous functionalities must continue to work correctly and results must be approved by the

supervisor.

5.10 Acceptance Testing

Acceptance testing ensures that the Sanad Platform satisfies the acceptance criteria and meets all client requirements.

This testing allows the customer to decide whether the system is ready for deployment.

Technique:

Real-life scenarios are performed, covering user roles, accessibility features, map interaction, bookings and admin tools.

Completion criteria:

All acceptance criteria must be fully met and the customer must approve the final system package.

5.11 Beta Testing

Beta testing is performed by selected end users using a pre-release version of the Sanad Platform to verify that it meets business and functional requirements.

The goal is to detect faults or issues that may appear only during real-world use.

Technique:

Users with disabilities, volunteers and administrators interact with the system and report any defects or usability issues.

Completion criteria:

All major defects must be resolved before final release.

6. PASS / FAIL CRITERIA

In this section, we describe the pass/fail criteria used in the Sanad System to ensure high reliability and reduce the chances of system failure. The system should be able to process user requests, validate digital services, and handle transactions without producing unexpected errors. A test case is considered passed when the system behavior matches the expected output and the functionality works according to the specifications.

A test case is considered failed if the system produces incorrect results, crashes, becomes unresponsive, or violates any of the functional or security requirements. Meeting these standards will ensure that Sanad provides users with a secure, stable, and user-friendly experience.

6.1 Suspension Criteria

Suspension criteria refer to situations in which the testing activities must be temporarily stopped. A suspension may apply to specific components or the entire system. Even if one module is suspended, other unrelated areas of the Sanad System may continue testing.

Testing is suspended in the following situations:

- When a critical failure occurs and prevents the continuation of test execution.
- When required external governmental services or APIs become unavailable.
- When inconsistent outputs appear for the same input across different modules.
- When the connection to the database or central authentication system fails.
- When testing or development is paused due to official holidays or system maintenance.

6.2 Resumption Criteria

Resumption criteria specify when suspended testing activities can be resumed. Testing may continue once the source of the issue has been resolved and stability is restored. Testing resumes when:

- Defect fixes are successfully applied and verified by the testing team.
- All external dependent services become available again.
- The testing environment is restored and confirmed to be functional.
- Database and authentication connections operate normally.
- Any scheduled maintenance or holiday-related downtime has ended.

6.3 Approval Criteria

Approval criteria ensure that the system is ready for release. A Sanad System component is considered approved only if:

- All test cases have been executed and passed.
- No high or critical defects remain unresolved.
- The actual results match the expected results in all test scenarios.
- The system demonstrates stable performance, security, and reliability.

Meeting these criteria means the testing phase is completed successfully and the system is ready for deployment.

7. TESTING PROCESS

This section outlines the deliverables, tasks, responsibilities, resources, and schedule necessary to complete the Sanad System testing phase.

7.1 Test Deliverables

The deliverables produced from the testing process include:

- Software Test Plan (STP)
- Test cases and test scripts
- Test input and output data
- Test execution logs
- Test incident/defect reports
- Test summary report
- Metrics and evaluation reports

After completing the testing phases and validating all requirements, the Sanad System will be prepared for deployment.

7.2 Testing Tasks

The Software Testing Plan is prepared after completing the Software Requirements Specification (SRS) and Software Design Specification (SDS). The main testing tasks include:

- Reviewing SRS and SDS documents
- Developing and organizing test cases
- Preparing software and hardware test environments
- Executing different types of testing such as functional, integration, and security testing
- Tracking defects and documenting errors found during execution

All testing activities and issues must be logged to maintain system quality.

7.3 Responsibilities

All team members are responsible for testing the Sanad System and verifying that all modules work correctly. Responsibilities include:

- Executing test cases
- Performing integration and functionality testing
- Documenting results and preparing reports
- Identifying and resolving issues that arise
- Delivering required testing documents

7.4 Resources

Resource Category	Description
Hardware	PC or laptop with stable internet connection
Software	• Windows / macOS
• MySQL Workbench	
• IntelliJ IDEA / NetBeans	
• Postman for API testing	
Support	None
Human	All team members based on their role and expertise

Table 20-Testing Resources

7.5 Schedule

Task	Date
Software requirement specification (SRS)	Nov. 10, 2025
Software design specification (SDS)	Nov. 24, 2025
Develop test cases	Dec. 3, 2025
Execute testing activities using various testing approaches	Dec. 5, 2025
Troubleshooting errors occurred during testing	Dec. 6, 2025
Modify the system accordingly	Dec. 7, 2025
Develop software testing plan (STP)	Dec. 8, 2025

Table 21-Testing Schedule

8. ENVIRONMENTAL REQUIREMENTS

In this section, the necessary hardware, software, communication facilities, security controls, tools and supporting documents for creating a conducive environment for executing all planned testing activities are described. A stable and reliable testing environment needs to be created with the ability for functional testing, integration testing, performance testing, accessibility testing and acceptance testing for supporting the Sanad platform.

8.1 Hardware

Hardware items necessary for the execution of test activities:

– Workstations:

Computers: laptops and desktops with:

- Intel Core i5 / AMD Ryzen 5 minimum processor
- At least 8 GB RAM (16 GB recommended)
- 256 GB SSD storage
- Standard Integrated Graphics

It ensures that the system can properly run development environment tools and browsers and supports testing tool integration.

-Mobile Devices:

- One Android device (Android 10 and later versions)
- One iOS device (iOS version 14 and later)

A mobile device is necessary for testing mobile accessibility and interaction across platforms.

– Network Connectivity:

- Stable high-speed internet connectivity

Needed for supporting cloud-based hosting and services and for handling APIs.

– Additional Facilities:

- Availability of a quiet laboratory or workspace
- Charging stations and ergonomic workspace design for prolonged testing activities

8.2 Software

A number of software applications must be used for test execution support:

- Operating Systems:

- Windows 10/11, macOS (supported versions), Android OS, and iOS

- Application and Testing Software:

- Web browsers: Google Chrome, Mozilla Firefox, Microsoft Edge

Tools for development and debugging:

- Visual Studio Code
- IntelliJ/WebStorm
- Database Tools: MySQL Workbench
- Backend testing tools: Node.js runtime
- Git version control service: GitHub
- Test tool for APIs: Postman
- Third-Party Integrations
- Google Maps Platform
- Video Communication Services
- Simulators for cloud deployment (for instance, AWS)

These integrations play a critical role in system components that utilize external services.

8.3 Security

The testing environment should support confidentiality, integrity and controlled access for test related data and system components.

The necessary conditions for accomplishing this include:

Enforcement of secure communication protocols (HTTPS)

- Employing anonymized test data and non-production databases

- Controlled access to testing servers and databases for authorized personnel only
- Storing and managing API keys and authentication tokens properly
- Verification of safeguarding from general attacks like SQL injection attacks, cross site scripting attacks, and incorrectly handled authentication attempts

These steps enable safe execution of test activities without compromising sensitive information.8.4 tools

8.4 Tools

Some specific tools and methods being used for testing purposes include:

- Functional Testing and UI Testing:

Selenium vs Cypress for Automated Web Interface Testing

Appium for mobile interface testing

- API Testing:

- Postman for testing endpoints and workflow validation

- Performance Testing:

- Apache JMeter for response time analysis and system capacity and stability tests

- Security Testing:

- OWASP ZAP and similar vulnerability scanners and threat analysis tools

- Accessibility Testing:

- Checking for accessibility standard compliance with WAVE, Axe DevTools, and the NVDA screen reader

Tool training is delivered through team-based workshops and self-learning modules.All analysis tool software is either open-source software or available under educational licenses.

8.5 Publications

Below are the necessary documents and references for supporting test planning and test execution:

- Software Requirements Specification (SRS)

- Software Design Specification (SDS)
- Software Project Management Plan (SPMP)

IEEE testing and documentation standards

- Accessibility guidelines (WCAG 2.1)
- Documentation for APIs and external services utilized by the platform

These publications facilitate the alignment of system design and behavior with test methods.

8.6 Risks and Assumptions

8.6.1 Risks:

- Unavailability of external services:
Third-party APIs might be down or rate limited for integration tests.

Mitigation: Use mock services during downtime.

- Device and Resource Limitations:

Inadequate access to mobile devices and cloud resources might hamper testing.

Mitigation: Share devices on schedules and use emulators when possible.

- Time constraints:

Deadlines for academic activities may limit the time available for testing.

Mitigation: Implement risk-based priorities for testing critical components.

- Frequent system modifications:

Changes to system logic and database design may make current test scenarios invalid.

Mitigation: It is important to version the schemes and test cases.

8.6.2 Assumptions:

- Reliable internet access can be expected throughout the testing process.
- The testing team would be able to access all necessary tools and test accounts.
- A production-like environment for testing is sufficient for validation.
- Testers will conduct testing with best practices for handling credentials and sensitive information.

9. CHANGE MANAGEMENT PROCEDURES

The Sanad Software Test Plan (STP) change management procedure is described in this section. It guarantees the consistent and traceable control, review, approval, and documentation of any changes made to the STP and associated testing artifacts.

9.1 Purpose

The procedures for change management are intended to:

- Throughout the Sanad project lifecycle, uphold the STP consistency and integrity.
- Make sure that every modification complies with the SDS and SRS documents.
- Prevent unapproved or unreviewed changes that could have a detrimental impact on the scope, schedule or quality of testing.

9.2 Items Under Change Control

The following items are subject to formal change control:

- Software Test Plan (STP) document (all versions).
- Test data sets, test scripts, and test cases.
- Test timetables and benchmarks.
- Configurations of the test environment (tools, versions, integrations).
- templates, checklists and reporting formats associated with testing.
- Every controlled item needs to be kept under the Sanad project space in a version-controlled repository (like GitHub or GitLab).

9.3 Change Initiation

When the STP or associated testing artifacts need to be updated, a change request (CR) is started because:

- New or modified requirements in the SRS or SDS.
- Discovery of gaps, ambiguities, or inconsistencies in the current STP.
- Changes in tools, environments, or external integrations.
- Feedback from testing activities (pilot run, beta testing, acceptance testing).
- Any of the following roles may propose a change:
 - Project Supervisor / Instructor.
 - Project Manager.
 - QA Lead.
 - Development Lead.
 - Accessibility Specialist or domain expert.

The requester must document the proposed change in a **Change Request Form**, including:

- Change ID.
- Date of request.
- Requester's name and role.
- Description of the change.
- Reason/justification.

9.4 Change Review and Impact Analysis

The Change Control Board (CCB) or other designated reviewers (such as the Project Supervisor, QA Lead, and Development Lead) examine the change request after it has been started.

During review, the following aspects must be analyzed:

- Impact on scope: Which features, modules, or test levels are impacted?
- Impact on the schedule: Will it be necessary to modify the testing schedule or milestones?
- Impact on resources: Does the change call for more environments, tools, or teamwork?
- Impact on risk: Does the modification create new risks or lessen current ones?
- Impact on traceability: Do links to SRS and SDS need to be updated or are they still valid?

The outcome of the review will be one of the following:

- **Approved with modifications**
- **Deferred** (to a future release or phase)
- **Rejected**

All review decisions must be recorded in the Change Request log

9.5 Change Authorization

A change is authorized only after formal approval by the responsible authority:

- For minor editorial or formatting changes: QA Lead approval is sufficient.
- For substantive changes that affect scope, schedule, or test coverage:
- Approval is required from the Project Supervisor and QA Lead, and may also involve the Project Manager or Development Lead, depending on impact.
- No change may be applied to the baseline STP or official test artifacts without documented authorization.

9.6 Implementation and Verification

After approval:

1. The responsible team member updates the relevant sections of the STP and/or test artifacts.
2. The updated documents are committed to the version control system with:
 - Clear commit message referencing the Change ID.
 - Updated version number (STP v1.0 → v1.1).
3. The QA Lead (or assigned reviewer) verifies that:
 - The change has been applied correctly.
 - Cross-references, tables, and numbering remain consistent.
 - No conflicts have been introduced with existing test plans or schedules.

If needed, selected test cases may be re-executed to validate that the change does not negatively impact ongoing testing.

9.7 Documentation, Versioning, and Communication

All approved changes must be:

- Logged in a Change Log section or a separate Change Management document.
- Included in the version history of the STP ("Version 1.1 – Updated Performance Testing approach").
- Communicated to all stakeholders (team members, supervisor, and other relevant parties) via email or project communication channels.

Stakeholders may obtain the latest approved version of the STP from the version control repository (GitHub/GitLab) at any time.

10. PLAN APPROVALS

Supervised by Dr. Enas El Shaarawy

#	Name	ID	Role
1	Khawla Hani Almoalem	2230003157	Leader
2	Fatimah Moayed Alawa	2230005350	Member
3	Zainab Esaeed	2230000664	Member
4	Zahra Aiman Alsadah	2230004418	Member
5	Jood Mohammed Alzayer	2230002118	Member
6	Fatimah Majed Almur	2230004016	Member
7	Remas Nasser Alqahtani	2230002281	Member